FACULTY OF APPLIED SCIENCES
UNIVERSITY OF WEST BOHEMIA
IN PILSEN

DEPARTMENT OF
COMPUTER SCIENCE
AND ENGINEERING

Seminar Work

Simulation of Electoral Systems

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Introduction 1

Electoral systems represent a key mechanism by which societies translate the individual preferences of voters into collective decisions. The choice of the right electoral system has a fundamental impact on election results, the fairness of representation, and the strategic behavior of voters. Despite the diversity of existing systems, none is universally optimal, and each exhibits specific characteristics that influence both the method of voting and the outcome of the election.

This work focuses on the mathematical and algorithmic analysis of various electoral systems, with the main goal being to simulate and compare their properties on the same set of voters and candidates. By modeling electoral processes, phenomena such as the spoiler effect or tactical voting can be examined. Simulations will allow quantifying the impacts of different vote conversion rules and show how different methods can lead to different results even with the exact same voter preferences.

The work first deals with the theoretical description of individual electoral systems and their properties. Subsequently, the simulation methodology is presented, which includes the choice of representation for voters and candidates, the implementation of voting strategies, and the rules for calculating results. Based on the simulation results, the advantages and disadvantages of individual approaches will be analyzed, and situations in which certain systems may fail or lead to paradoxical results will be discussed.

Electoral Systems 2

An electoral system is, in short, the way a group of people makes a democratic decision. A common and well-known area of use is in politics for electing representatives to lead a country. There are many systems that can be used for this. For example, in the Czech Republic, at least 2 different systems are used. A multi-round majoritarian plurality system is used for the presidential election, and a combined proportional system with a plurality system is used in elections to the Chamber of Deputies. Other systems include, for example, the alternative vote (instant runoff) or approval voting. Each system has its pros and cons, but none is completely fair.

2.1 Majoritarian vs. Proportional Electoral System

Electoral systems can generally be divided into majoritarian and proportional. A majoritarian electoral system is characterized by the fact that the candidate who wins the majority wins it all, and the winner takes all. The most typical examples include the absolute majority system and the relative majority system.

A proportional electoral system is specific in that the win is proportionally divided among the first n-candidates. An example in the Czech Republic is the election to the Chamber of Deputies, where the final number of mandates is proportionally divided among parties that obtained at least 5%.

Furthermore, this work will only deal with majoritarian electoral systems because the first part of the electoral process is the same in both types of systems and they only differ when evaluating votes and redistributing the win. In a majoritarian system, the distribution of the win does not need to be addressed - the winner takes all.

2.2 Examples of Majoritarian Electoral Systems

2.2.1 Relative Majority System (Plurality)

The relative majority system (also called first-past-the-post) is a single-round electoral system where voters can cast exactly one vote, and the candidate who receives the highest number of votes wins. [Ver]

Such a system tempts voters to use tactics if their favorite does not have a high chance of winning. Let's imagine an election with 2 strong candidates, A and B, and one weaker candidate, C. Candidate A has about a 46% chance of winning, candidate B about 44%, and candidate C about 10%. If everyone voted for their favorite, candidate A would win. But the majority of candidate C's voters prefer candidate B over A. Therefore, if at least 3% of the voters voted for their second choice - candidate B - instead of candidate C, candidate B would win. This phenomenon is called the "Spoiler effect" [Pri].

2.2.2 Multi-round Absolute Majority System (Plurality runoff)

The absolute majority system consists in the fact that a candidate must obtain an absolute majority of votes (more than 50% of all votes) to win. It is most often used in the form of a multi-round electoral system, where in each round the candidates with the lowest number of votes are eliminated until one of the candidates gets more than 50% of the votes. The use of two rounds is typical, where voters cast exactly one vote in the first round, and if neither candidate gains an absolute (over half) majority of votes, the two most successful candidates with the highest number of votes advance to the second round. Whoever gets more votes out of the pair in the second round becomes the winner. If there is a tie between two candidates in the second round, which in practice almost never happens with a large number of votes, the winner can be decided by another supplementary technique - e.g., by drawing lots, calling another round or new elections, etc.

Compared to the relative majority system, this system allows voters to vote honestly in the first round for their favorite regardless of their chances of winning.

2.2.3 Alternative Vote (Instant runoff)

In the alternative vote, the voter ranks (numbers) the candidates according to their preferences. First, it is determined whether any candidate obtained more than 50% of the first preferences (votes). If so, they become the winner. If not, the candidate with the lowest number of first preferences is eliminated. Their votes are transferred

to the candidates who were in the next (in this case, second) place in the ranking on the eliminated ballots. This procedure is repeated until one of the candidates has an absolute majority of votes. [Gre]

The alternative vote can be used in several variants, namely in combination with compulsory voting (the voter must give preferences to all candidates), optional voting (the voter can give as many preferences as they want, even if it is just one preference/one vote), or limited voting (compared to previous variants, the voter must give an amount of votes in a narrower interval between one vote and all votes).

This electoral system in its various forms allows voters to express their preferences honestly without tactical voting. The voter can honestly choose their favorite because even if they drop out due to the lowest count, their vote will not be lost but will be counted towards the voter's second choice.

But even such a system has its flaw. Let's imagine situation number 1. We have 3 candidates A, B, and C. Candidate A gets 25% of the votes, candidate B 30%, and candidate C 45%. No one obtained over 50%, so the candidate with the lowest number of votes will be eliminated and their votes redistributed to their second choice. However, since candidates A and C have very different views, all of candidate A's voters chose candidate B as their second option - so candidate B will win with 55% of the votes. Situation number 2: But if candidate C were not strong enough and some of their voters switched to candidate A, the results could look like this: candidate A - 31%, candidate B - 30%, and candidate C - 39%. In such a situation, candidate B would be eliminated first, and their votes would be redistributed to their second choice. Candidate B was opinion-wise somewhere between candidates A and C, and half the voters put candidate A as their second choice and the other half put candidate C. After redistribution, candidate A will have 46% of the votes and candidate C - 54%. Candidate C would therefore win even though they were weaker than in the first situation, where they did not win.

2.2.4 Approval Voting

In approval voting, voters can cast a vote for any number of candidates, where each approved candidate receives exactly one vote from the voter. The winner here is the candidate approved by the largest number of voters. [H21]

This system is resistant to phenomena such as the "Spoiler effect" or the described undesirable phenomenon in the alternative vote. However, it does not allow voters to express a preference among candidates, only to evaluate whether yes or no.

This can be solved by a simple modification of the voting, where instead of voting yes or no for each candidate, the voter can rate each candidate. This expresses how much they prefer or dislike a given candidate. The rating can be, for example, in the range of -10 to 10 or 0 to 5, where the lowest number means - I do not support

them at all, I do not want them to win, and the highest number means the highest level of support, they are my favorite. However, this results in both a more complex system to understand and to evaluate the votes. At the same time, this can lead to large differences between more and less popular candidates.

2.2.5 Condorcet Method

The Condorcet method is an electoral system in which the candidate who would win against each of the opposing candidates in a head-to-head vote wins. This condition is called the Condorcet criterion, and a candidate elected this way is the Condorcet winner. Such a winner most fairly represents the votes of the voters. [PCb]

There are several ways to implement the Condorcet method, but they all lead to the same result. These methods are the fairest but have one major negative: a Condorcet winner may not always exist.

In a hypothetical situation where we have 3 candidates (A, B, C) and 3 voters (1, 2, 3), each voter is supposed to rank the candidates according to their preferences. The voters rank their candidates as follows:

Voter 1: A > B > C; Voter 2: C > A > B; Voter 3: B > C > A

In a vote between candidates A and B, candidate A would win, as they are preferred over B by voters 1 and 2, which is the majority. Between candidates B and C, candidate B would win, as they are preferred by voters 1 and 3. And in a vote between A and C, candidate C would win because voters 2 and 3 prefer them. Therefore, no candidate would win against every single opponent, and a Condorcet winner does not exist.

The probability of such a situation occurring decreases as the number of voters increases, but it never decreases to 0%.

Simulation of an Electoral System

3

There are a number of ways to simulate an electoral system, varying in their fidelity to reality. The level of fidelity is influenced both by the way the voter and candidate are represented, and by the amount of external and internal influences the simulation allows to be simulated - e.g., the publication of election predictions can affect the voter's final decision.

3.1 Representation of the Voter and Candidate

The basic element of any such simulation is the representation of the voter and candidate, and the decision of which candidates individual voters will vote for. Probably the simplest way is representation using a one-dimensional axis, on which voters and candidates are placed. The voter then gives their vote to the closest candidate or candidates. The axis can represent, for example, the left and right wings or any 2 political directions. Voters and candidates are then placed on the axis according to their inclination to given directions.

Slightly more complex is a representation by a two-dimensional space (2D plane). The principle is the same as for a one-dimensional space, but instead of 2 political directions, 4 can be taken into account. In this way, it is possible to add more dimensions as needed, which, however, significantly complicates visualization.

A new experimental representation is a representation by a binary array of length n, which represents the opinion of the voter/candidate on n questions - a yes/no answer to a question. The voter then compares their opinion matches with candidates, resulting in a certain percentage match with each. This is how, for example, online election calculators work. This method of representation is not used in election simulators, and it will be tested and observed in this work.

A possible modification is the use of an integer array of size n. This allows the voter to rate the extent to which they hold or reject a given opinion.

3.1.1 Candidate Popularity

Candidate popularity is a subjective concept and has no clear definition. In this case, popularity will represent the candidate's strength, persuasiveness, and trust. It will serve to estimate the extent of their reach and their ability to convince a citizen to vote. Popularity will be stronger among aligned voters than among other voters. At the same time, it will be used to estimate whether a voter will use tactical voting and decide to support a less preferred candidate so that their least favorite one does not win.

3.2 Simulation of the Electoral Process

The voting process itself is simple. First, voters vote, then all their votes are counted, and subsequently, they are evaluated. The simulation of such a process is not demanding and follows reality. [PCa]

1. First, the preferences of each voter are evaluated, based on which the voter ranks all candidates by popularity.
2. The voter votes according to the created ranking of candidates.
3. The votes are then counted and evaluated.

Such voting would be completely ideal because it would exactly match what the citizens desire. Unfortunately, this is not how it works in reality. Not every citizen goes to vote, and not every voter decides to vote according to their preference (voting for their preferred candidate), but they use tactical voting and vote for the lesser evil. The voter does not want a strong candidate they dislike to win, so they vote for a less preferred candidate who has a better chance of winning.

Whether a citizen goes to vote can be implemented using the popularity of their preferred candidates. The more popular a candidate is, the more citizens they convince to vote. And whether a voter will vote for their preferred candidate can be implemented using the popularity and the chance of winning of their least preferred candidates.

Simulation Goals 4

The goal of this simulation is to demonstrate various electoral systems and observe how the results change with different systems, but with the exact same voters and candidates. At the same time, the goal is to observe and describe phenomena that can occur in the given electoral systems and to decide whether they are desirable.

The first step will be to create a set of voters and candidates who will be placed in a two-dimensional space. The second step will be the implementation of the mentioned electoral systems and the visualization of the results of individual elections. Furthermore, interesting phenomena that can occur in the given systems will be observed.

Finally, the work will deal with the implementation and observation of voter representation using a binary and integer array of size n, which represent the voter's agreement with opinions, and candidate popularity. Both methods are proposed by the author and it will be tested how they affect the simulation.

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